

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 9 — The Periodic Table: Chemical Periodicity

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	9.1 Periodicity of Physical Properties (Period 3) · 9.2 Periodicity of Chemical Properties (Oxides & Chlorides)
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	235 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

9.1 Physical Periodicity (Period 3)

— Trends across Period 3: atomic radius, ionic radius, melting points, electrical conductivity, and such

9.2 Chemical Periodicity (Oxides & Chlorides)

— Reactions of Period 3 elements with O₂, Cl₂, and H₂O; structure, bonding, acid-base character of c

SUBTOPIC 9.1 — PHYSICAL PERIODICITY ACROSS PERIOD 3

1 Electronic Configurations Across Period 3 — 9701/22/M/J/21/Q8(a)–(b)

EASY · Syllabus 9.1

The physical and chemical properties of the Period 3 elements (Na to Ar) are fundamentally determined by their electronic configurations.

- (a) Write the full electronic configuration of an atom of silicon, Si, and an atom of sulfur, S. [2]

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- (b) State the number of unpaired electrons present in the ground state of an isolated phosphorus atom, P. [1]

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2 Physical Property Trends Across Period 3 — 9701/21/O/N/20/Q7(a)–(d)

HARD · Syllabus 9.1

Fig. 9.1 displays the variation of atomic radius, first ionisation energy, melting point, and electrical conductivity across Period 3.

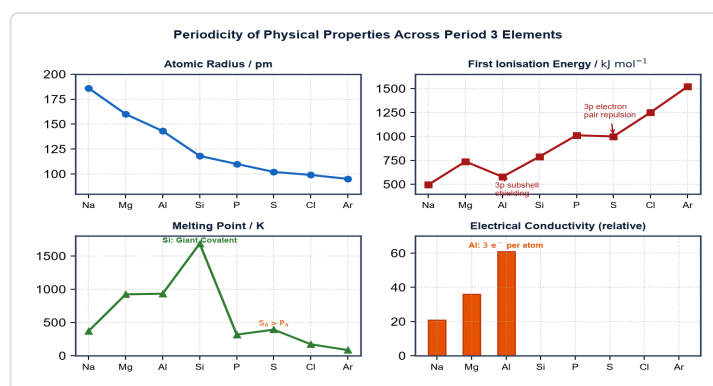


Fig. 9.1: Trends in atomic radius, first ionisation energy, melting point, and electrical conductivity across Period 3.

- (a) Explain why atomic radius decreases steadily across Period 3 from Na to Ar. [2]

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- (b) Explain why the first ionisation energy of aluminium is lower than that of magnesium. [2]

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- (c) Explain why silicon has by far the highest melting point of all elements in Period 3. [2]

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- (d) Explain why electrical conductivity increases from Na to Al, drops sharply at Si, and is virtually zero from P to Ar. [3]

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3 Ionic Radii Across Period 3: Cations vs Anions — 9701/23/M/J/22/Q8(a)–(c)

HARD · Syllabus 9.1

Fig. 9.2 shows the variation of ionic radius across Period 3.

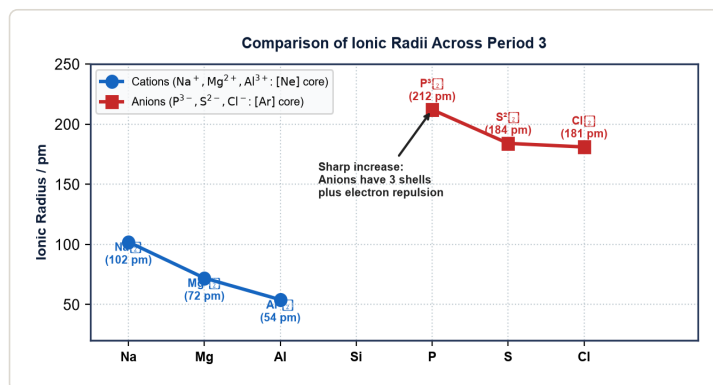


Fig. 9.2: Comparison of ionic radii across Period 3 showing cation and anion series.

- (a) Explain why the ionic radius decreases across the cation series: Na⁺ (102 pm) > Mg²⁺ (72 pm) > Al³⁺ (54 pm). [2]

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- (b) Explain the massive discontinuity / sudden increase in ionic radius between Al³⁺ (54 pm) and P³⁻ (212 pm). [3]

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- (c) Explain why the ionic radius decreases across the anion series: P³⁻ (212 pm) > S²⁻ (184 pm) > Cl⁻ (181 pm). [2]

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4 Trend in Atomic Radius: Nuclear Charge vs Shielding — 9701/22/F/M/21/Q8(a)–(b)

EASY · Syllabus 9.1

Across Period 3 from sodium (186 pm) to chlorine (99 pm), the atomic radius decreases by nearly 50%.

- (a) Identify the two competing factors that influence atomic radius across a period, and state which factor predominates. [2]

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- (b) Explain why electron shielding remains approximately constant across Period 3. [1]

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5 First Ionisation Energy Dip at Sulfur — 9701/21/M/J/22/Q8(a)–(b)

HARD · Syllabus 9.1

The first ionisation energy of phosphorus is 1012 kJ mol⁻¹, whereas that of sulfur is 1000 kJ mol⁻¹, contrary to the general upward trend.

- (a) Draw orbital box diagrams for the 3p subshells of phosphorus and sulfur. [2]

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- (b) Explain the dip in first ionisation energy between phosphorus and sulfur. [2]

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6 Melting Point Trends of Period 3 Non-Metals — 9701/22/O/N/23/Q8(a)–(c)

HARD · Syllabus 9.1

The melting points of the non-metallic elements in Period 3 are:

- Silicon: 1414 °C
- Phosphorus: 44 °C
- Sulfur: 115 °C
- Chlorine: –101 °C
- Argon: –189 °C

- (a) State the molecular formula of elemental phosphorus, sulfur, and chlorine in their standard states. [3]

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- (b) Explain why sulfur has a significantly higher melting point than phosphorus. [2]

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- (c) Explain why argon has the lowest melting point of all Period 3 elements. [2]

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7 Electrical Conductivity of Period 3 Elements — 9701/21/M/J/23/Q8(a)–(b)

EASY · Syllabus 9.1

Electrical conductivity varies widely across Period 3 elements.

- (a) Explain why aluminium is a better electrical conductor than sodium. [2]

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- (b) Explain why solid sulfur is an electrical insulator. [1]

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8 Deducing Group Number from Successive Ionisation Energies of Silicon — 9701/22/M/J/20/Q8(a)–(b)

EASY · Syllabus 9.1

The first six successive ionisation energies of an unknown Period 3 element X are:

789, 1577, 3232, 4356, 16091, 19785 kJ mol^{–1}.

- (a) Identify element X and justify your choice by locating the major jump in ionisation energy. [2]

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- (b) Write the equation for the fifth ionisation energy of element X. [1]

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9 Radius of Sodium Cation vs Sodium Atom — 9701/21/O/N/22/Q8(a)-(b)

EASY · Syllabus 9.1

The radius of a neutral sodium atom is 186 pm, whereas the radius of a sodium cation, Na^+ , is 102 pm.

- (a) State the electron configuration of Na and Na^+ . [1]

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- (b) Explain why the Na^+ cation is significantly smaller than the Na atom. [2]

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10 Radius of Chloride Anion vs Chlorine Atom — 9701/23/O/N/21/Q7(a)-(b)

EASY · Syllabus 9.1

The atomic radius of chlorine is 99 pm, whereas the ionic radius of the chloride ion, Cl^- , is 181 pm.

- (a) Explain why the chloride anion is nearly twice as large as the neutral chlorine atom. [2]

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- (b) State whether this expansion upon ionization is characteristic of all non-metal anions. [1]

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11 Isoelectronic Cations: Na^+ , Mg^{2+} , Al^{3+} — 9701/22/F/M/23/Q8(a)-(b)

HARD · Syllabus 9.1

The ions Na^+ , Mg^{2+} , and Al^{3+} are isoelectronic with neon.

- (a) Define the term isoelectronic. [1]

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- (b) Arrange these three ions in order of decreasing ionic radius and explain the trend. [3]

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12 Isoelectronic Anions: P³⁻, S²⁻, Cl⁻ — 9701/21/M/J/21/Q9(a)–(b)

HARD · Syllabus 9.1

The ions P³⁻, S²⁻, and Cl⁻ all possess 18 electrons (1s² 2s² 2p⁶ 3s² 3p⁶).

- (a)** Explain why P³⁻ has the largest ionic radius and Cl⁻ has the smallest among these three ions. **[2]**

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- (b)** Compare the ionic radius of Cl⁻ (181 pm) with that of K⁺ (138 pm), both of which possess 18 electrons. **[2]**

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13 Pauling Electronegativity Trend Across Period 3 — 9701/22/M/J/22/Q8(a)–(b)

EASY · Syllabus 9.1

Electronegativity values increase steadily across Period 3: Na (0.9), Mg (1.2), Al (1.5), Si (1.8), P (2.1), S (2.5), Cl (3.0).

- (a)** Define electronegativity. **[2]**

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- (b)** Explain why electronegativity increases across Period 3. **[2]**

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14 Density Variations Across Period 3 — 9701/22/O/N/21/Q9(a)–(b)

HARD · Syllabus 9.1

The densities of Period 3 elements are:

Na (0.97), Mg (1.74), Al (2.70), Si (2.33), P (1.82), S (2.07), Cl (0.003), Ar (0.0017 g cm⁻³).

- (a)** Explain why density increases from sodium to aluminium. **[2]**

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- (b)** Explain the massive drop in density between sulfur and chlorine at room temperature. **[2]**

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15 Second Ionisation Energy Anomaly: Sodium vs Magnesium — 9701/21/O/N/23/Q8(a)–(b)

HARD · Syllabus 9.1

The first and second ionisation energies of sodium and magnesium are:

• Sodium: IE₁ = 496 kJ mol⁻¹, IE₂ = 4562 kJ mol⁻¹

• Magnesium: IE₁ = 738 kJ mol⁻¹, IE₂ = 1451 kJ mol⁻¹

- (a)** Explain why the second ionisation energy of sodium is more than three times greater than that of magnesium. **[3]**

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- (b) Write the equation for the second ionisation energy of sodium. [1]

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16 Metallic Bond Strength in Period 3 Metals — 9701/23/M/J/20/Q7(a)–(b)

EASY · Syllabus 9.1

The boiling points of the Period 3 metals increase in the order: Na (883 °C) < Mg (1090 °C) < Al (2470 °C).

- (a) Describe the bonding in a metal. [2]

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- (b) Explain why the metallic bond strength increases from sodium to aluminium. [2]

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17 Silicon as a Semiconductor — 9701/22/F/M/24/Q7(a)–(b)

HARD · Syllabus 9.1

Silicon is a metalloid element located between the metallic conductors and non-metallic insulators in Period 3.

- (a) Contrast the effect of increasing temperature on the electrical conductivity of silicon with that of copper metal. [2]

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- (b) Explain your answer to (a) at the particulate level. [2]

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18 Third Ionisation Energy of Magnesium — 9701/21/M/J/24/Q9(a)–(b)

EASY · Syllabus 9.1

For magnesium, the first three ionisation energies are:

IE1 = 738, IE2 = 1451, IE3 = 7733 kJ mol⁻¹.

- (a) Explain why IE3 is more than five times larger than IE2 for magnesium. [2]

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- (b) Predict whether aluminium or magnesium has the higher third ionisation energy. Justify your answer. [2]

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19 Bond Enthalpy in Homonuclear Period 3 Molecules — 9701/22/O/N/24/Q8(b)

HARD · Syllabus 9.1

Consider the covalent bonding in Cl₂, P₄, and S₈.

- (a)** State the number of single covalent bonds present in one molecule of P₄ and one molecule of S₈. **[2]**

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- (b)** Explain why white phosphorus, P₄, is tetrahedral and highly strained, causing it to ignite spontaneously in air. **[2]**

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20 Discontinuity in Ionisation Energy Between Mg and Al — 9701/21/O/N/24/Q8(a)–(b)

HARD · Syllabus 9.1

Across Period 3, first ionisation energies generally increase. However, $IE_1(\text{Mg}) = 738 \text{ kJ mol}^{-1}$ while $IE_1(\text{Al}) = 578 \text{ kJ mol}^{-1}$.

- (a)** Give the subshell notation for the electron removed in Mg and in Al. **[2]**

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- (b)** Explain how this provides experimental evidence for subshell electronic structure. **[2]**

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21 Physical State and Volatility Trends Across Period 3 — 9701/23/M/J/23/Q7(a)–(b)

EASY · Syllabus 9.1

The elements in Period 3 span metals, metalloids, non-metal solids, and gases at 20 °C.

- (a)** State which two elements in Period 3 are gases at 20 °C and standard atmospheric pressure. **[1]**

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- (b)** Explain why chlorine is a gas at room temperature while iodine is a solid, in terms of intermolecular forces. **[2]**

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22 First Ionisation Energy of Argon vs Potassium — 9701/22/M/J/25/Q7(a)–(b)

HARD · Syllabus 9.1

The first ionisation energy of argon (Period 3) is 1521 kJ mol^{-1} , whereas that of potassium (Period 4) is 419 kJ mol^{-1} .

- (a)** Explain the dramatic drop in first ionisation energy from argon to potassium. **[3]**

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- (b)** State the period and group of potassium. **[1]**

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23 Periodicity of Molar Volume Across Period 3 — 9701/21/F/M/25/Q8(a)–(b)

EASY · Syllabus 9.1

Molar volume is defined as atomic mass divided by density ($V_m = A_r / \rho$).

- (a) Explain why the molar volume of sodium ($23.7 \text{ cm}^3 \text{ mol}^{-1}$) is larger than that of magnesium ($14.0 \text{ cm}^3 \text{ mol}^{-1}$). [2]

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- (b) State how the strength of metallic bonding influences the packing density and molar volume. [1]

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24 Group 2 vs Group 1 Periodic Trends — 9701/22/O/N/25/Q8(b)

EASY · Syllabus 9.1

Comparing sodium (Group 1) with magnesium (Group 2) in Period 3 illustrates key group differences.

- (a) Compare the hardness and melting points of sodium and magnesium. [2]

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- (b) Explain these differences by referring to their metallic lattices. [2]

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25 Thermal Conductivity Trends Across Period 3 — 9701/23/O/N/24/Q7(a)–(b)

EASY · Syllabus 9.1

Metals in Period 3 are excellent thermal conductors, whereas non-metals are thermal insulators.

- (a) Describe the mechanism of thermal conduction in metallic aluminium. [2]

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- (b) Explain why solid phosphorus and sulfur are very poor thermal conductors. [1]

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SUBTOPIC 9.2 — CHEMICAL PERIODICITY: OXIDES & CHLORIDES

26 Period 3 Oxides: Acid-Base Continuum — 9701/22/M/J/21/Q9(a)–(d)

HARD · Syllabus 9.2

Fig. 9.3 illustrates the progressive change in acid-base character of Period 3 oxides from basic to amphoteric to acidic.

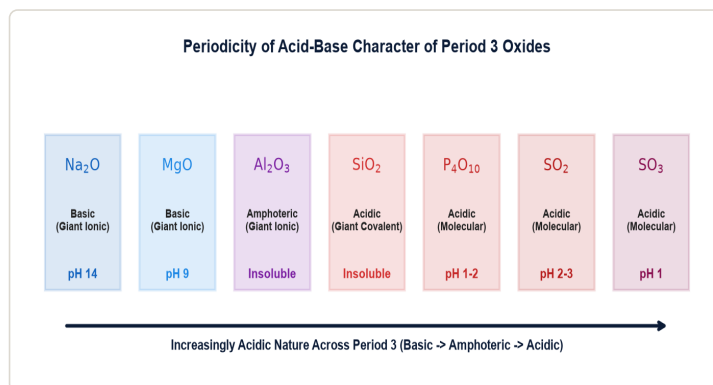


Fig. 9.3: Periodicity of acid-base behavior of Period 3 oxides.

- (a) State the structure and bonding in Na_2O , Al_2O_3 , and SO_3 . [3]
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- (b) Write the equation for the reaction of sodium oxide with water and state the pH of the resulting solution. [2]
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- (c) Explain what is meant by an amphoteric oxide, using aluminium oxide, Al_2O_3 , as an example with two equations. [3]
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- (d) Write the equation for the reaction of phosphorus(V) oxide, P_4O_{10} , with water and state the approximate pH. [2]
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Period 3 Chlorides: Structure, Hydrolysis & Solution pH — 9701/21/O/N/21/Q8(a)-(d)

HARD · Syllabus 9.2

Fig. 9.4 summarizes the bonding, structure, and behavior with water for Period 3 chlorides.

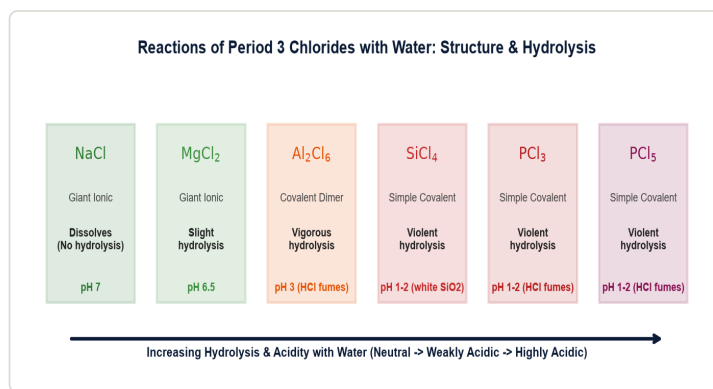


Fig. 9.4: Comparison of structure, hydrolysis, and pH of Period 3 chlorides in water.

- (a) Describe the difference in observations when water is added to solid NaCl compared to liquid SiCl_4 . [2]
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- (b) Write the equation for the reaction of SiCl_4 with water, including state symbols. [2]

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- (c) Explain why aqueous sodium chloride has a neutral pH of 7 while aqueous aluminium chloride is distinctly acidic (pH ~ 3). [3]

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- (d) Write the equation for the reaction of phosphorus(V) chloride, PCl_5 , with an excess of water. [2]

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28 Combustion of Period 3 Elements in Oxygen — 9701/23/M/J/22/Q9(a)–(c)

EASY · Syllabus 9.2

Period 3 elements react vigorously with oxygen gas upon heating.

- (a) State the flame colour observed when each of the following elements burns in oxygen: [3]

- (i) Sodium
(ii) Magnesium
(iii) Sulfur

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- (b) Write balanced chemical equations for the combustion of sodium (forming Na_2O) and sulfur (forming SO_2). [2]

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29 Reactions of Period 3 Elements with Chlorine Gas — 9701/22/F/M/22/Q7(a)–(b)

EASY · Syllabus 9.2

Chlorine gas is a strong oxidising agent that reacts with Period 3 elements to form chlorides.

- (a) Write the balanced chemical equation for the reaction of heated aluminium with dry chlorine gas. [2]

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- (b) Describe the structure and bonding in the dimeric molecule Al_2Cl_6 formed in the vapour phase. [2]

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30 Reactions of Sodium and Magnesium with Water — 9701/21/M/J/22/Q9(a)–(c)

HARD · Syllabus 9.2

Sodium and magnesium exhibit contrasting reactivities with water.

- (a) Describe three observations when a small piece of sodium is dropped into cold water. [3]

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- (b) Write the chemical equation for the reaction of magnesium with cold water and state the approximate pH of the resulting mixture. [2]

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- (c) Write the chemical equation for the reaction of heated magnesium with steam and state two observations. [3]

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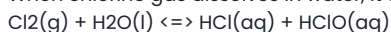
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31 Disproportionation of Chlorine in Water — 9701/22/O/N/22/Q7(a)–(c)

HARD · Syllabus 9.2

When chlorine gas dissolves in water, it undergoes a reversible disproportionation reaction:



- (a) Deduce the oxidation state of chlorine in Cl_2 , HCl , and HClO . [3]

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- (b) Explain why damp blue litmus paper turns red and then rapidly bleaches white when exposed to chlorine gas. [2]

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- (c) State the active chemical species responsible for the antibacterial disinfection of drinking water. [1]

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32 Solubility and Basicity of Na_2O vs MgO — 9701/23/O/N/22/Q7(a)–(b)

HARD · Syllabus 9.2

Both sodium oxide, Na_2O , and magnesium oxide, MgO , are basic ionic oxides.

- (a) Explain why Na_2O dissolves completely in water to produce a solution of pH 14, whereas MgO produces a suspension of pH 9. [3]

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- (b) Write the ionic equation for the reaction of oxide ions, O^{2-} , with water. [1]

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33 Insoluble Oxides: Al_2O_3 and SiO_2 — 9701/22/F/M/23/Q9(a)–(b)

EASY · Syllabus 9.2

When aluminium oxide and silicon(IV) oxide are added to water, no reaction is observed and the pH remains 7.

- (a) Explain why Al_2O_3 does not dissolve in water in terms of lattice energy. [2]

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- (b) Explain why SiO_2 does not dissolve in water in terms of its macromolecular structure. [2]

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34 Reactions of Acidic Oxides with Water — 9701/21/M/J/23/Q9(a)–(c)

EASY · Syllabus 9.2

The non-metallic oxides P_4O_{10} , SO_2 , and SO_3 are acidic and react with water.

- (a) Write the balanced equation for the reaction of sulfur dioxide, SO_2 , with water and state the formula of the acid formed. [2]

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- (b) Write the balanced equation for the reaction of sulfur trioxide, SO_3 , with water and state the formula of the acid formed. [2]

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- (c) Write the balanced equation for the reaction of phosphorus(V) oxide with water. [1]

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35 Amphoteric Nature of Aluminium Oxide — 9701/22/M/J/23/Q7(a)–(b)

HARD · Syllabus 9.2

Aluminium oxide, Al_2O_3 , exhibits dual chemical behaviour by reacting with both strong acids and strong alkalis.

- (a) Write the ionic equation for the reaction of Al_2O_3 with dilute sulfuric acid. [2]

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- (b) Write the ionic equation for the reaction of Al_2O_3 with hot concentrated aqueous sodium hydroxide. [2]

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36 Hydrolysis of Aluminium Chloride Dimer, Al_2Cl_6 — 9701/21/O/N/23/Q9(a)–(c)

HARD · Syllabus 9.2

Solid aluminium chloride consists of Al_2Cl_6 dimers. When water is added dropwise, steamy fumes and an acidic solution are formed.

- (a) Identify the gas responsible for the steamy acidic fumes. [1]

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- (b) In aqueous solution, aluminium ions form the hexaaqua complex ion $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$. Write an equation showing how this complex ion hydrolyses water to produce an acidic solution. [2]
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- (c) Explain why $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ hydrolyses water whereas $[\text{Na}(\text{H}_2\text{O})_6]^+$ does not. [2]
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37 Violent Hydrolysis of Silicon(IV) Chloride — 9701/22/O/N/23/Q9(b)

EASY · Syllabus 9.2

Silicon(IV) chloride, SiCl_4 , is a volatile liquid at room temperature that fumes profusely in moist air.

- (a) Describe two observations when liquid SiCl_4 is added to water. [2]
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- (b) Write the balanced equation for the complete hydrolysis of SiCl_4 . [2]
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38 Hydrolysis of Phosphorus Chlorides: PCl_3 vs PCl_5 — 9701/23/M/J/24/Q9(a)–(c)

HARD · Syllabus 9.2

Both phosphorus(III) chloride and phosphorus(V) chloride react vigorously with water to form acidic solutions.

- (a) Write the balanced equation for the reaction of PCl_3 with water and deduce the oxidation state of phosphorus in the product acid. [3]
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- (b) Write the balanced equation for the reaction of PCl_5 with water and deduce the oxidation state of phosphorus in the product acid. [3]
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39 Acid-Base Reactions of Period 3 Oxides with Sulfuric Acid — 9701/21/M/J/24/Q10(a)–(b)

EASY · Syllabus 9.2

Basic and amphoteric oxides react with acids to form salts and water.

- (a) Write the equation for the reaction of solid magnesium oxide with dilute sulfuric acid. [2]
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- (b) Write the equation for the reaction of solid aluminium oxide with dilute sulfuric acid. [2]
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40 Reaction of Acidic Oxides with Alkalis: SiO₂ and SO₂ — 9701/22/M/J/24/Q10(a)–(b)**HARD · Syllabus 9.2**

Acidic oxides neutralise alkalis to form oxoanion salts.

- (a) Silicon(IV) oxide does not react with aqueous NaOH at room temperature. Write the equation for its reaction with hot, concentrated molten sodium hydroxide. [2]

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- (b) Write the equation for the neutralisation of sulfur dioxide by aqueous sodium hydroxide. [2]

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41 Phosphate Fertilizer Synthesis from P₄O₁₀ — 9701/23/O/N/23/Q8(a)–(b)**HARD · Syllabus 9.2**

Phosphorus(V) oxide reacts with basic calcium oxide at high temperatures to form calcium phosphate, Ca₃(PO₄)₂, a primary constituent of phosphate fertilizers.

- (a) Write the balanced equation for this solid-state acid-base synthesis. [2]

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- (b) State which reactant acts as the acid and which reactant acts as the base. [1]

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42 Sulfur Oxides and Environmental Acid Rain — 9701/22/F/M/25/Q9(a)–(c)**HARD · Syllabus 9.2**

The combustion of sulfur-containing fossil fuels produces sulfur dioxide, which converts to acid rain:

- Stage 1: $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$
- Stage 2: $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$
- Stage 3: $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$

- (a) State two damaging environmental consequences of acid rain. [2]

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- (b) Explain why rainwater in pristine unpolluted air is naturally slightly acidic with a pH of ~5.6 rather than 7.0. [2]

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43 Contrast: Covalent Al₂Cl₆ vs Ionic Al₂O₃ in Water — 9701/21/F/M/25/Q9(a)–(b)**HARD · Syllabus 9.2**

Aluminium forms both a chloride, Al₂Cl₆, and an oxide, Al₂O₃.

- (a) Explain why Al₂Cl₆ reacts vigorously with water whereas Al₂O₃ is completely insoluble. [3]

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- (b) Deduce the coordination number of aluminium in Al_2Cl_6 vapour and in $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$. [2]

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44 Corrosion Resistance & Passivation of Aluminium — 9701/22/M/J/25/Q9(a)–(b)

EASY · Syllabus 9.2

Although aluminium is a highly electropositive metal with a strongly negative standard electrode potential (-1.66 V), aluminium drinks cans and window frames do not react with water or air.

- (a) Explain why aluminium appears chemically unreactive under ambient conditions. [2]

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- (b) Explain what happens when this protective layer is disrupted by adding aqueous mercury(II) chloride or sodium hydroxide. [2]

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45 Deducing an Unknown Period 3 Element from Compound Properties — 9701/23/M/J/25/Q8(a)–(c)

HARD · Syllabus 9.2

An unknown Period 3 element E forms a solid oxide that dissolves in water to give an acidic solution of pH 1. Element E also forms a volatile liquid chloride that reacts violently with water to give steamy acidic fumes and an acidic solution.

- (a) Deduce the identity of element E. [1]

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- (b) Write the formula of the oxide and the chloride formed by element E in its highest oxidation state. [2]

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- (c) Write the equation for the reaction of the chloride of E with excess water. [2]

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46 Highest Oxidation States Across Period 3 — 9701/21/O/N/25/Q9(a)–(b)

EASY · Syllabus 9.2

The maximum oxidation state exhibited by Period 3 elements increases across the period to match the group number.

- (a) State the maximum oxidation state of each element from Na to Cl in its highest oxide: [4]
Na, Mg, Al, Si, P, S, Cl

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- (b) Explain why argon forms no oxides or chlorides under standard conditions. [1]

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Why CCl₄ Resists Hydrolysis While SiCl₄ Hydrolyses Violently — 9701/22/O/N/25/Q9(a)–(c)

HARD · Syllabus 9.2

Both tetrachloromethane, CCl₄, and silicon(IV) chloride, SiCl₄, are non-polar tetrahedral covalent liquids. When shaken with water, CCl₄ forms an immiscible layer with zero reaction, whereas SiCl₄ hydrolyses violently.

- (a) Explain why SiCl₄ undergoes rapid hydrolysis while CCl₄ is completely inert to water. [3]

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- (b) Explain the role of vacant 3d orbitals in the silicon atom during this mechanism. [2]

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48

Flame Emission of Period 3 Elements — 9701/23/O/N/25/Q8(a)–(b)

EASY · Syllabus 9.2

Flame emission spectra arise from electronic transitions.

- (a) Explain the atomic origin of the distinctive yellow-orange flame produced by sodium compounds. [3]

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- (b) Explain why magnesium does not produce a characteristic flame colour in a standard Bunsen flame, but burns with blinding white light. [2]

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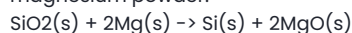
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49

Redox Reduction of Silica by Magnesium — 9701/21/M/J/25/Q9(a)–(b)

HARD · Syllabus 9.2

Elemental silicon can be prepared in the laboratory by the vigorous exothermic redox reaction between silicon(IV) oxide and magnesium powder:



- (a) Identify the oxidising agent and reducing agent in this reaction, stating the changes in oxidation numbers. [3]

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- (b) Describe how pure silicon can be separated from the product mixture containing MgO, Si, and unreacted Mg. [2]

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Comprehensive Synoptic: Period 3 Periodicity, Oxide Acidity & Chloride Hydrolysis — 9701/22/O/N/25/Q9(a)–(d) **ASD · Syllabus 9.2**

Table 9.1 summarises data for the elements from sodium to sulfur across Period 3:

- Na: giant metallic, basic oxide Na_2O , chloride NaCl (pH 7 in water)
- Al: giant metallic, amphoteric oxide Al_2O_3 , chloride Al_2Cl_6 (pH 3 in water)
- Si: giant covalent, acidic oxide SiO_2 , chloride SiCl_4 (pH 1 in water)
- P: simple molecular, acidic oxide P_4O_{10} , chloride PCl_5 (pH 1 in water)

- (a) Explain the trend in bonding and structure of the elements from Na to P.

[3]

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- (b) Explain why the acid-base character of the oxides changes from strongly basic (Na_2O) to amphoteric (Al_2O_3) to acidic (SiO_2 , P_4O_{10}).

[3]

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- (c) Explain the variation in pH when the chlorides NaCl , Al_2Cl_6 , and SiCl_4 are separately added to water.

[3]

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- (d) Write the equation for the reaction of amphoteric Al_2O_3 with aqueous sodium hydroxide.

[1]

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Mark Scheme — Topic 9 — The Periodic Table: Chemical Periodicity
CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	Si: 1s2 2s2 2p6 3s2 3p2 (1); S: 1s2 2s2 2p6 3s2 3p4 (1)	2
1(b)	3 unpaired electrons (one in each of 3px, 3py, 3pz by Hund's rule) (1)	1
2(a)	Nuclear charge (number of protons) increases across the period while electron shielding remains approximately constant (electrons added to same 3rd shell) (1); Stronger electrostatic attraction pulls valence electrons closer to the nucleus (1)	2
2(b)	The outer electron in Al is in a 3p orbital, which is at a higher energy level than the 3s orbital of Mg (1); The 3p electron is shielded by the inner 3s2 electrons, requiring less energy to remove (1)	2
2(c)	Silicon has a giant covalent 3D lattice structure (diamond-like macromolecule) (1); Melting requires breaking millions of strong covalent Si-Si bonds throughout the entire structure, requiring huge thermal energy (1)	2
2(d)	From Na to Al, metallic bonding has more delocalised electrons per atom (1 in Na, 2 in Mg, 3 in Al), increasing mobile charge carrier density (1); Si is a giant covalent metalloid where valence electrons are localised in covalent bonds with only few thermally excited electrons (semiconductor) (1); P to Ar are simple molecular or monoatomic non-metals with all electrons localised in bonds/lone pairs, so no mobile charge carriers exist (1)	3
3(a)	Na+, Mg2+, and Al3+ are isoelectronic with identical [Ne] core (10 electrons in 2 shells) (1); Nuclear charge increases (+11 to +12 to +13), pulling the remaining electron shells more tightly towards the nucleus (1)	2
3(b)	Cations (Na+ to Al3+) have lost their outer 3rd shell and have only 2 electron shells (1); Anions (P3- to Cl-) retain 3 electron shells (isoelectronic with [Ar], 18 electrons) (1); Gaining extra electrons causes mutual electron-electron repulsion, pushing the electron cloud outward (1)	3
3(c)	P3-, S2-, and Cl- are isoelectronic with 18 electrons (3 shells) (1); Nuclear charge increases (+15 to +16 to +17), pulling the same number of electrons into a progressively smaller volume (1)	2
4(a)	Increasing nuclear charge (proton number) and electron shielding (1); Increasing nuclear charge predominates across the period (1)	2
4(b)	All successive electrons are added into the same principal quantum shell (n = 3), with the same number of inner shielding electrons (1s2 2s2 2p6) (1)	1
5(a)	P: three single arrows in separate boxes [^][^][^] (1); S: one pair of arrows and two single arrows [^v][^][^] (1)	2
5(b)	In sulfur, the electron is removed from a paired 3p orbital (3p_x^2) (1); Spin-pair repulsion between the two electrons in the same orbital makes that electron easier to remove than from an unpaired orbital in P (1)	2
6(a)	Phosphorus: P4 (1); Sulfur: S8 (1); Chlorine: Cl2 (1)	3
6(b)	Both have simple molecular structures; S8 molecules are larger and possess more electrons (128 electrons) than P4 molecules (60 electrons) (1); Stronger London dispersion forces between S8 molecules require more thermal energy to overcome (1)	2
6(c)	Argon exists as monoatomic atoms (Ar) with no covalent molecular structure (1); It has the smallest electron cloud and weakest London dispersion forces, requiring minimal energy to separate (1)	2
7(a)	Aluminium contributes 3 delocalised electrons per atom to the metallic lattice compared to only 1 for sodium (1); Aluminium has a much higher concentration of mobile charge carriers (electrons) per unit volume (1)	2
7(b)	In sulfur (S8), all valence electrons are fixed/localised in covalent bonds or lone pairs; there are no delocalised electrons or mobile ions to carry current (1)	1
8(a)	Element X is Silicon / Si (1); The huge jump occurs between the 4th and 5th IE (4356 to 16091 kJ mol^-1), indicating that 4 electrons are in the outer shell and the 5th is removed from an inner noble gas shell (Group 14) (1)	2
8(b)	Si^4+(g) -> Si^5+(g) + e- (1)	1
9(a)	Na: 1s2 2s2 2p6 3s1; Na+: 1s2 2s2 2p6 (1)	1
9(b)	Formation of Na+ involves the loss of the entire outermost quantum shell (n = 3) (1); In Na+, 11 protons now pull on only 10 electrons, increasing effective nuclear charge per electron and pulling remaining shells closer (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
10(a)	Adding an extra electron increases electron-electron repulsion in the outer 3p subshell (1); The nuclear charge (+17) is unchanged, so the electron cloud expands outward to minimize repulsions (1)	2
10(b)	Yes, all negative ions (anions) are larger than their corresponding neutral parent atoms (1)	1
11(a)	Having the same number of electrons / identical electronic configuration (1)	1
11(b)	Na^+ (102 pm) > Mg^{2+} (72 pm) > Al^{3+} (54 pm) (1); All three have 10 electrons in two shells with identical shielding (1); Proton number increases: Na (11) < Mg (12) < Al (13), creating greater nuclear attraction that pulls electrons closer (1)	3
12(a)	Nuclear charge increases from P (+15) to S (+16) to Cl (+17) (1); The 17 protons in Cl^- exert a stronger attractive force on the 18 electrons than the 15 protons in P^{3-} , pulling shells tighter (1)	2
12(b)	K^+ has 19 protons while Cl^- has only 17 protons (1); Greater nuclear pull in K^+ draws the 18 electrons in closer, making K^+ smaller than Cl^- (1)	2
13(a)	The power / ability of an atom to attract the shared pair of electrons (1); in a covalent bond (1)	2
13(b)	Nuclear charge increases while shielding remains roughly constant (1); Atomic radius decreases, so the bonded electron pair is closer to the nucleus and attracted more strongly (1)	2
14(a)	Atomic mass increases (23 to 24.3 to 27) while atomic volume decreases (atoms pack closer together due to stronger metallic bonding) (1); Mass per unit volume increases (1)	2
14(b)	Sulfur is a solid with molecules packed in a crystalline lattice (1); Chlorine is a gas at room temperature with molecules separated by vast empty spaces (1)	2
15(a)	For sodium, the second electron is removed from an inner 2p orbital (noble gas core [Ne]) (1); This electron is closer to the nucleus and experiences much less shielding (1); For magnesium, the second electron is removed from the same 3s valence orbital, experiencing shielding from the full $n = 2$ core (1)	3
15(b)	$\text{Na}^+(\text{g}) \rightarrow \text{Na}^{2+}(\text{g}) + \text{e}^-$ (1)	1
16(a)	Electrostatic attraction between a regular lattice of positive metal ions (cations) (1); and a sea of delocalised electrons (1)	2
16(b)	Ionic charge increases (Na^+ to Mg^{2+} to Al^{3+}) and ionic radius decreases (1); Number of delocalised electrons per atom increases (1 to 2 to 3), giving much stronger electrostatic attraction (1)	2
17(a)	In silicon, electrical conductivity increases with temperature (1); In copper, electrical conductivity decreases with temperature (1)	2
17(b)	In silicon, thermal energy promotes electrons across the small band gap from valence into conduction band (1); In copper, higher temperature causes metal ions to vibrate more vigorously, scattering conducting electrons and increasing resistance (1)	2
18(a)	The third electron is removed from an inner 2p orbital (noble gas core) (1); It is much closer to the nucleus and experiences far less electron shielding (1)	2
18(b)	Magnesium has the higher IE ₃ (1); In Mg, the 3rd electron is from inner 2p shell, whereas in Al (Group 13), the 3rd electron is still a valence electron from the 3s orbital (IE ₃ (Al) ~ 2745 kJ mol ⁻¹) (1)	2
19(a)	In P ₄ : 6 single P-P bonds (tetrahedron of 4 atoms) (1); In S ₈ : 8 single S-S bonds (crown-shaped ring) (1)	2
19(b)	In P ₄ , P-P-P bond angles are constrained to 60° within the triangular faces (1); This is far smaller than the preferred 109.5° tetrahedral angle, creating severe ring strain that lowers activation energy for combustion (1)	2
20(a)	Mg: 3s electron (from 3s ²) (1); Al: 3p electron (from 3s ² 3p ¹) (1)	2
20(b)	The lower IE of Al despite having higher nuclear charge (+13 vs +12) proves that the 3p subshell is at a higher energy level than the 3s subshell (1); and that the 3s electron pair partially shields the 3p electron (1)	2
21(a)	Chlorine (Cl ₂) and Argon (Ar) (1)	1
21(b)	Iodine molecules (I ₂) have far more electrons than chlorine molecules (Cl ₂) (106 vs 34 electrons) (1); Iodine electron cloud is more polarisable, leading to significantly stronger London dispersion forces that hold molecules in a solid crystal at room temperature (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
22(a)	The outer electron in potassium enters a new quantum shell (4s) further from the nucleus (1); The 4s electron experiences significantly greater shielding from the full 3s ² 3p ⁶ inner noble gas core (1); Increased distance and shielding far outweigh the increase in nuclear charge (+19 vs +18), making the 4s electron much easier to remove (1)	3
22(b)	Period 4, Group 1 (1)	1
23(a)	Sodium has a larger atomic radius than magnesium (186 vs 160 pm) and weaker metallic bonding (1); Sodium atoms are held less tightly and pack less densely (1)	2
23(b)	Stronger metallic bonding pulls cations closer together, reducing molar volume and increasing density (1)	1
24(a)	Sodium is soft (can be cut with a knife) with low mp (98 °C); Magnesium is hard with high mp (650 °C) (2)	2
24(b)	Magnesium ions have a 2+ charge and smaller radius compared to Na ⁺ (1); Magnesium has double the density of delocalised electrons, creating vastly stronger metallic bonds that resist mechanical deformation and thermal cleavage (1)	2
25(a)	Thermal energy increases kinetic vibrations of Al ³⁺ ions which are passed to neighbouring ions (1); Mobile delocalised electrons absorb kinetic energy and rapidly diffuse through the metal, transferring heat quickly (1)	2
25(b)	They lack delocalised mobile electrons, and molecules are separated by weak intermolecular forces that transmit lattice vibrations poorly (1)	1
26(a)	Na ₂ O: Giant ionic (1); Al ₂ O ₃ : Giant ionic with covalent character (1); SO ₃ : Simple covalent molecular (1)	3
26(b)	Na ₂ O(s) + H ₂ O(l) → 2NaOH(aq) (1); pH ~ 13-14 (strongly alkaline) (1)	2
26(c)	An oxide that reacts with both acids and bases/alkalis to form salts (1); Acid: Al ₂ O ₃ + 6HCl → 2AlCl ₃ + 3H ₂ O (1); Base: Al ₂ O ₃ + 2NaOH + 3H ₂ O → 2NaAl(OH) ₄ (or Al ₂ O ₃ + 2OH ⁻ → 2AlO ₂ ⁻ + H ₂ O) (1)	3
26(d)	P ₄ O ₁₀ (s) + 6H ₂ O(l) → 4H ₃ PO ₄ (aq) (1); pH ~ 1-2 (strongly acidic) (1)	2
27(a)	NaCl dissolves quietly to form a clear, colourless solution with no temperature rise (1); SiCl ₄ reacts violently with water, producing steamy white fumes of HCl gas and a white precipitate of hydrated SiO ₂ (1)	2
27(b)	SiCl ₄ (l) + 2H ₂ O(l) → SiO ₂ (s) + 4HCl(aq) (or SiCl ₄ + 4H ₂ O → Si(OH) ₄ + 4HCl) (2)	2
27(c)	Na ⁺ has low charge density (+1, large radius) and does not polarise coordinated water molecules (1); Al ³⁺ has a very high charge density (+3, small radius) and polarises O-H bonds in [Al(H ₂ O) ₆] ³⁺ (1); [Al(H ₂ O) ₆] ³⁺ hydrolyses water, releasing H ⁺ ions: [Al(H ₂ O) ₆] ³⁺ ⇌ [Al(H ₂ O) ₅ (OH)] ²⁺ + H ⁺ (1)	3
27(d)	PCl ₅ (s) + 4H ₂ O(l) → H ₃ PO ₄ (aq) + 5HCl(aq) (2)	2
28(a)	(i) Bright yellow / orange-yellow flame (1); (ii) Brilliant blinding white flame (1); (iii) Blue flame (1)	3
28(b)	4Na(s) + O ₂ (g) → 2Na ₂ O(s) (1); S(s) + O ₂ (g) → SO ₂ (g) (1)	2
29(a)	2Al(s) + 3Cl ₂ (g) → Al ₂ Cl ₆ (s) (or 2Al + 3Cl ₂ → 2AlCl ₃) (2)	2
29(b)	Two AlCl ₃ units linked together by two coordinate (dative covalent) bonds (1); Each dative bond is formed by a lone pair of electrons from a chlorine atom donating into the vacant 3p orbital of an aluminium atom (1)	2
30(a)	Melts into a silvery ball / bead (1); Fizzes / effervesces vigorously and darts across water surface (1); Leaves an alkaline solution (universal indicator turns purple) / shrinks and disappears (1)	3
30(b)	Mg(s) + 2H ₂ O(l) → Mg(OH) ₂ (s) + H ₂ (g) (1); pH ~ 9-10 (weakly alkaline due to low solubility of Mg(OH) ₂) (1)	2
30(c)	Mg(s) + H ₂ O(g) → MgO(s) + H ₂ (g) (1); Bright white light / blinding white flame (1); White solid / powder of MgO formed (1)	3
31(a)	In Cl ₂ : 0 (1); In HCl: -1 (1); In HClO: +1 (1)	3
31(b)	Turns red due to presence of acidic H ⁺ ions from HCl and HClO (1); Then bleaches white because chloric(I) acid (HClO) is a powerful oxidising agent that bleaches dyes (1)	2
31(c)	Chloric(I) acid, HClO (or chlorate(I) ion, ClO ⁻) (1)	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
32(a)	Na ₂ O has a relatively low lattice energy (+1 ions) and dissolves readily to form fully dissociated Na ⁺ and OH ⁻ ions, giving high [OH ⁻] and pH 14 (1); MgO has an extremely high lattice energy (Mg ²⁺ and O ²⁻ ions carry 2+ and 2- charges), making it sparingly soluble in water (1); Only a tiny concentration of OH ⁻ is produced at equilibrium, yielding pH ~ 9 (1)	3
32(b)	O ²⁻ (s) + H ₂ O(l) → 2OH ⁻ (aq) (1)	1
33(a)	Al ³⁺ and O ²⁻ ions carry high charges (3+ and 2-), resulting in an immense lattice energy (1); The hydration energy of the ions is insufficient to overcome this very strong electrostatic lattice (1)	2
33(b)	SiO ₂ has a giant covalent 3D tetrahedral network (1); Water molecules cannot break the very strong covalent Si-O bonds throughout the macromolecule (1)	2
34(a)	SO ₂ (g) + H ₂ O(l) ⇌ H ₂ SO ₃ (aq) (1); Sulfurous acid / sulfuric(IV) acid, H ₂ SO ₃ (1)	2
34(b)	SO ₃ (g) + H ₂ O(l) → H ₂ SO ₄ (aq) (1); Sulfuric acid / sulfuric(VI) acid, H ₂ SO ₄ (1)	2
34(c)	P ₄ O ₁₀ (s) + 6H ₂ O(l) → 4H ₃ PO ₄ (aq) (1)	1
35(a)	Al ₂ O ₃ (s) + 6H ⁺ (aq) → 2Al ³⁺ (aq) + 3H ₂ O(l) (2)	2
35(b)	Al ₂ O ₃ (s) + 2OH ⁻ (aq) + 3H ₂ O(l) → 2[Al(OH) ₄] ⁻ (aq) (or Al ₂ O ₃ + 2OH ⁻ → 2AlO ₂ ⁻ + H ₂ O) (2)	2
36(a)	Hydrogen chloride / HCl(g) (1)	1
36(b)	[Al(H ₂ O) ₆] ³⁺ (aq) + H ₂ O(l) ⇌ [Al(H ₂ O) ₅ (OH)] ²⁺ (aq) + H ₃ O ⁺ (aq) (or [Al(H ₂ O) ₆] ³⁺ ⇌ [Al(H ₂ O) ₅ (OH)] ²⁺ + H ⁺) (2)	2
36(c)	Al ³⁺ has a small ionic radius and high charge (+3), giving an extremely high charge density (1); It strongly polarises coordinated water molecules, weakening O-H bonds and releasing H ⁺ ions, whereas Na ⁺ has low charge density (+1) and cannot polarise water (1)	2
37(a)	Vigorous / violent reaction with evolution of dense white steamy fumes of HCl (1); Formation of a white gelatinous precipitate of SiO ₂ / hydrated silicon dioxide (1)	2
37(b)	SiCl ₄ (l) + 2H ₂ O(l) → SiO ₂ (s) + 4HCl(g) (or SiCl ₄ + 4H ₂ O → Si(OH) ₄ + 4HCl) (2)	2
38(a)	PCl ₃ (l) + 3H ₂ O(l) → H ₃ PO ₃ (aq) + 3HCl(aq) (2); Phosphorus in H ₃ PO ₃ has oxidation state +3 (no change) (1)	3
38(b)	PCl ₅ (s) + 4H ₂ O(l) → H ₃ PO ₄ (aq) + 5HCl(aq) (2); Phosphorus in H ₃ PO ₄ has oxidation state +5 (no change) (1)	3
39(a)	MgO(s) + H ₂ SO ₄ (aq) → MgSO ₄ (aq) + H ₂ O(l) (2)	2
39(b)	Al ₂ O ₃ (s) + 3H ₂ SO ₄ (aq) → Al ₂ (SO ₄) ₃ (aq) + 3H ₂ O(l) (2)	2
40(a)	SiO ₂ (s) + 2NaOH(l) → Na ₂ SiO ₃ (l) + H ₂ O(g) (2)	2
40(b)	SO ₂ (g) + 2NaOH(aq) → Na ₂ SO ₃ (aq) + H ₂ O(l) (or SO ₂ + NaOH → NaHSO ₃) (2)	2
41(a)	P ₄ O ₁₀ (s) + 6CaO(s) → 2Ca ₃ (PO ₄) ₂ (s) (2)	2
41(b)	Acid: P ₄ O ₁₀ (acidic oxide); Base: CaO (basic oxide) (1)	1
42(a)	Corrosion / chemical weathering of limestone and marble buildings / monuments (CaCO ₃) (1); Acidification of lakes and rivers, killing fish and aquatic life / leaching toxic Al ³⁺ ions from soil that damage tree roots (1)	2
42(b)	Naturally occurring atmospheric carbon dioxide, CO ₂ , dissolves in rainwater (1); forming weak, partially dissociated carbonic acid: CO ₂ + H ₂ O ⇌ H ₂ CO ₃ ⇌ H ⁺ + HCO ₃ ⁻ (1)	2
43(a)	Al ₂ Cl ₆ consists of discrete covalent molecules held by weak intermolecular forces; water molecules readily attack the vacant d-orbitals on Al, releasing immense hydration energy and hydrolysing the weak covalent Al-Cl bonds (1); Al ₂ O ₃ has an extremely strong giant ionic lattice with high lattice energy (Al ³⁺ and O ²⁻ ions) which water cannot overcome (1); Hydration enthalpy cannot compensate for the immense lattice energy (1)	3
43(b)	In Al ₂ Cl ₆ : 4 (tetrahedral around each Al) (1); In [Al(H ₂ O) ₆] ³⁺ : 6 (octahedral) (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
44(a)	Aluminium reacts spontaneously with atmospheric oxygen to form an extremely thin, impervious, strongly adhering layer of aluminium oxide, Al ₂ O ₃ (1); This passivation layer physically seals the underlying metal from contact with water or oxygen (1)	2
44(b)	The oxide layer is dissolved or breached, exposing fresh metallic aluminium (1); The bare aluminium reacts vigorously and exothermically with water / oxygen (1)	2
45(a)	Phosphorus / P (1)	1
45(b)	Oxide: P ₄ O ₁₀ (or P ₂ O ₅) (1); Chloride: PCl ₅ (or PCl ₃) (1)	2
45(c)	PCl ₅ + 4H ₂ O → H ₃ PO ₄ + 5HCl (2)	2
46(a)	Na: +1, Mg: +2, Al: +3, Si: +4 (2); P: +5, S: +6, Cl: +7 (2)	4
46(b)	Argon has a complete outer octet (3s ² 3p ⁶) with extremely high ionisation energy and zero electron affinity; it is thermodynamically stable as isolated monoatomic atoms (1)	1
47(a)	Silicon has accessible, energetically low-lying vacant 3d orbitals in its valence shell, allowing it to expand its octet (1); Water molecules can donate a lone pair into an empty 3d orbital on Si to form an activated 5-coordinate intermediate (1); Carbon has only 2s and 2p orbitals (maximum octet of 8 electrons) with no available d-orbitals; carbon is also sterically hindered by 4 bulky chlorine atoms (1)	3
47(b)	Vacant 3d orbitals accept a lone pair of electrons from a nucleophilic H ₂ O molecule (1); enabling a transition state with expanded coordination number (e.g. SiCl ₄ (H ₂ O)) before expelling a Cl ⁻ ion (1)	2
48(a)	Thermal energy from the flame excites valence electrons from the 3s ground state to higher energy 3p orbitals (1); When excited electrons drop back to the lower 3s energy level, they emit electromagnetic radiation (1); The energy gap corresponds to photons of wavelength ~589 nm in the visible yellow region (ΔE = h × ν) (1)	3
48(b)	Thermal energy in a standard Bunsen flame is insufficient to excite electrons in Mg (higher ionisation energy) (1); When burning vigorously, the extreme combustion temperature heats solid MgO particles to incandescence, emitting continuous white light (blackbody radiation) (1)	2
49(a)	Reducing agent: Magnesium / Mg (0 to +2) (1); Oxidising agent: Silicon / Si in SiO ₂ (+4 to 0) (1); Correct assignment of oxidation states (1)	3
49(b)	Add excess dilute hydrochloric acid (1); Basic MgO and unreacted Mg dissolve to form aqueous MgCl ₂ and H ₂ , while silicon is insoluble in acid and can be filtered off, washed, and dried (1)	2
50(a)	Na and Al have giant metallic lattices with cations in delocalised electrons (1); Si has a giant covalent 3D macromolecular network with localised shared pairs (1); P has a simple molecular lattice (P ₄) with weak London dispersion forces between discrete molecules (1)	3
50(b)	Electronegativity of Period 3 elements increases across the period (1); Oxides transition from ionic lattices with free basic O ²⁻ ions (Na ₂ O) to partially covalent with polarised bonds (Al ₂ O ₃) to fully covalent molecular/macromolecular species where polarized non-metal atoms attract OH ⁻ or donate H ⁺ to water (SiO ₂ , P ₄ O ₁₀) (2)	3
50(c)	NaCl simply dissolves into hydrated Na ⁺ and Cl ⁻ without hydrolysing water, maintaining neutral pH 7 (1); Al ₂ Cl ₆ forms [Al(H ₂ O) ₆] ³⁺ whose high charge density polarises coordinated water, releasing H ⁺ to give pH ~ 3 (1); SiCl ₄ undergoes violent covalent hydrolysis, generating free HCl(aq) that fully dissociates into H ⁺ and Cl ⁻ , giving pH ~ 1 (1)	3
50(d)	Al ₂ O ₃ (s) + 2NaOH(aq) + 3H ₂ O(l) → 2NaAl(OH) ₄ (aq) (1)	1